Homework 3: RAG for Science Question Answering

RNN and Transformer Course

Spring 2026

Abstract

This report presents the official Homework 3 pipeline using a 50-question random sample from the specified Kaggle `train.csv` file with `random_state=42`. The system compares two chunking strategies, builds two FAISS indices with `BAAI/bge-m3`, applies two-stage retrieval with dense search followed by `cross-encoder/ms-marco-MiniLM-L-6-v2`, and generates final answers with local Ollama `llama3`. On the official evaluation set, the final generation pipeline reached **70.00% accuracy (35/50)**. A previous science-only / DeepSeek / ensemble branch is retained only as an optional appendix and is not used as the formal HW3 score.

Contents

1 Introduction

The goal of Homework 3 is to build a retrieval-augmented generation pipeline for science question answering. The official submission in this report follows the required structure:

- Part 1: compare two chunking strategies and build two vector indices.
- Part 2: implement dense retrieval plus cross-encoder re-ranking and compare hit rate.
- Part 3: use Ollama Llama-3 to answer multiple-choice questions from retrieved evidence.
- Latency analysis: measure vector search, re-ranking, and generation time on the same official pipeline.

2 Dataset and Official Evaluation Setup

2.1 Random 50-question sample from train.csv

The official evaluation set is created directly from the specified Kaggle train file:

```
train\_path = r"D:\\course\\rnnlstm\\HW3\\kaggle-llm-science-exam\\train.csv"
train\_df = pd.read\_csv(train\_path)
official\_eval\_df = train\_df.sample(n=50, random\_state=42).reset\_index(drop=True)
official\_eval\_df = official\_eval\_df.rename(columns={"prompt": "question"})
```

This produces a fixed 50-question sample from the 200 training rows in the specified file. This is the only dataset used for the formal HW3 results in this report.

2.2 Wikipedia corpus construction

The cached `official_wiki_articles.json` is used as the local Wikipedia plain-text corpus for this assignment run. Articles are converted into document objects before chunking.

Metric	Value
specified `train.csv` rows	200
Official evaluation rows	50
Raw Wikipedia documents	500
Total raw corpus characters	2,252,107

3 Part 1: Indexing Pipeline

3.1 Chunking Strategy A

Strategy A uses fixed-size recursive chunking with `chunk_size=500` and `overlap=50`. This is the smaller-granularity baseline index.

3.2 Chunking Strategy B

Strategy B uses larger recursive chunks with `chunk_size=1000` and `overlap=200`. This keeps more local context inside each chunk.

3.3 Vector Index Construction

Both indices use the same embedding model and GPU execution:

- Embedding model: `BAAI/bge-m3`
- Vector store: FAISS
- Device: CUDA GPU

Index A stores Strategy A chunks, and Index B stores Strategy B chunks.

3.4 Chunk Size Analysis

Metric	Strategy A	Strategy B
Chunk size	500	1000
Overlap	50	200
Number of chunks	6,865	3,204
Average chunk length (chars)	334.48	732.73
Maximum chunk length (chars)	500	1000

Strategy A creates many shorter chunks, while Strategy B produces fewer but longer chunks. In this rerun, smaller chunks more frequently cut sentence boundaries and split complete answer spans across chunk edges, while Strategy B preserved longer contiguous evidence and gave more reliable complete-answer capture.

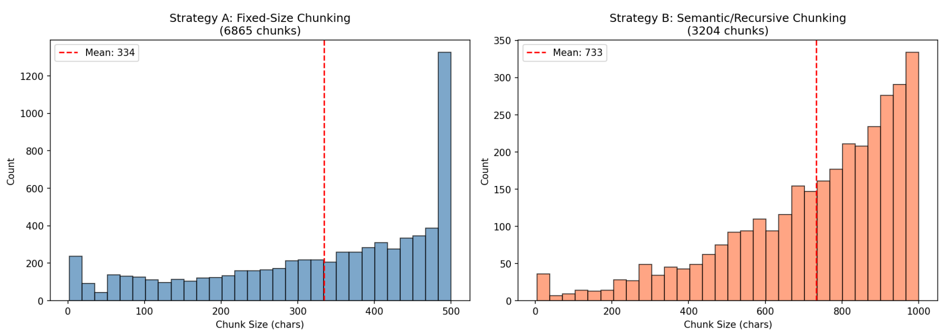

Figure 1: Chunk-size distribution from the notebook run.

4 Part 2: Retrieval and Re-ranking

4.1 Dense Retrieval

Stage 1 retrieval uses dense vector search over each FAISS index with `top_k=20`.

4.2 Cross-Encoder Re-ranking

Stage 2 uses `cross-encoder/ms-marco-MiniLM-L-6-v2` to score the 20 retrieved candidates and keep the top 3 chunks for generation.

4.3 Hit Rate Comparison

Hit rate is reported as a retrieval proxy metric based on keyword overlap between the retrieved chunks and the correct answer text. The assignment requires this comparison, so it is kept explicitly as a proxy retrieval metric rather than a perfect relevance label.

Index	Method	Hit Rate
Index A (fixed 500/50)	Vector Search Only	66.00%
Index A (fixed 500/50)	Vector Search + Re-ranking	72.00%
Index B (recursive 1000/200)	Vector Search Only	78.00%
Index B (recursive 1000/200)	Vector Search + Re-ranking	78.00%

On this official 50-question sample, Strategy B gives the strongest vector-only hit-rate proxy, while the final official generation pipeline uses Strategy B with re-ranking as the required two-stage retrieval setup. Re-ranking improves Index A and ties Index B on this proxy table; therefore we report gains carefully as retrieval-proxy improvements and still rely on end-to-end QA accuracy for final assessment.

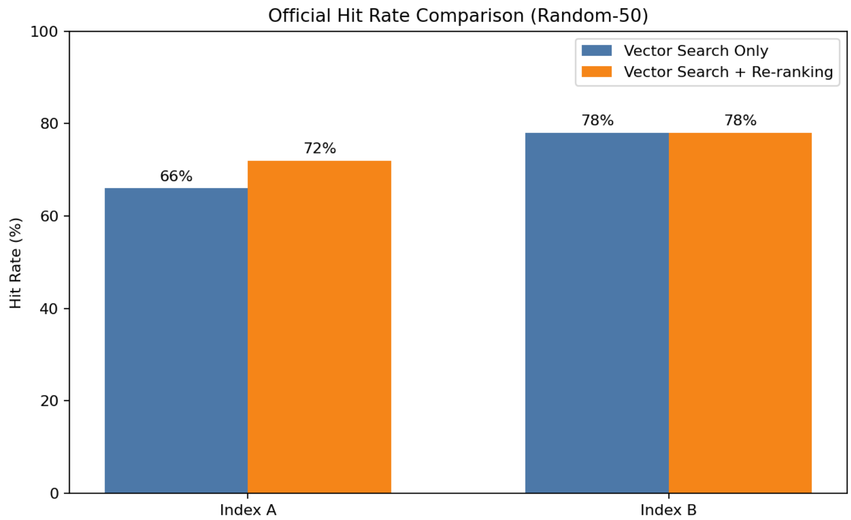

Figure 2: Official hit-rate comparison from the notebook run.

4.4 Two Qualitative Re-ranking Examples

The following two cases were selected from the official random-50 evaluation set. They show cases where vector search ranked a weak or irrelevant chunk first, while cross-encoder re-ranking promoted a more answer-relevant chunk to rank 1.

Case 1: MOND concept promotion

- Source: official random-50

- Question: What is Modified Newtonian Dynamics (MOND)?
- Correct answer: `C. MOND is a hypothesis that proposes a modification of Newton's law of universal gravitation ...`
- Vector Search top-1:
- snippet: `Magnetic field`
- score: `-10.3605`
- why it is weak / irrelevant: the vector top-1 chunk is off-topic and does not explain MOND or modified gravity.
- Re-ranked top-1:
- snippet: `Sir Isaac Newton's laws of motion contain an important formula ... one newton ... acceleration due to earth's gravity ...`
- original vector rank: `6`
- score: `-6.9466`
- evidence phrase: `Newton law of universal gravitation`
- why it is more answer-relevant / supports the correct answer: the promoted chunk brings in Newton-law gravity concepts, which are the core concept needed to interpret MOND as a modification framework.
- Explanation: cross-encoder re-ranking moved an irrelevant top-1 chunk down and promoted a gravity-concept chunk to rank 1 with a clear score gain.

Case 2: Light-ion fusion concept promotion

- Source: official random-50
- Question: What is accelerator-based light-ion fusion?
- Correct answer: `A. Accelerator-based light-ion fusion is a technique that uses particle accelerators ... to induce light-ion fusion reactions ...`
- Vector Search top-1:
- snippet: `Photon interactions with matter`
- score: `-9.7298`
- why it is weak / irrelevant: the vector top-1 chunk is about generic photon interactions and does not explain accelerator-based fusion setup.
- Re-ranked top-1:
- snippet: `The light and heat created by nuclear fusion makes stars like the Sun very hot ... Fusion makes a lot of energy ...`
- original vector rank: `9`

- score: `-5.8277`
- evidence phrase: `light and heat created by nuclear fusion`
- why it is more answer-relevant / supports the correct answer: the promoted chunk directly discusses fusion-process evidence, which is far closer to the correct option than the vector top-1 chunk.
- Explanation: cross-encoder re-ranking promoted a fusion-bearing chunk from rank 9 to rank 1 and improved the relevance score.

These examples are qualitative evidence-ordering demonstrations and do not change the official random-50 accuracy computation.

5 Part 3: Ollama Generation

5.1 Llama-3 Integration

The official generator is local Ollama `llama3`.

5.2 Anti-hallucination Prompt

The system prompt enforces two constraints:

- use only the retrieved context,
- output exactly one letter from `A/B/C/D/E`.

For open-ended RAG usage, the same policy maps insufficient evidence to "I do not know", while this HW3 MCQ run enforces single-letter output.

The generator receives the reranked top-3 chunks together with the question and answer choices.

5.3 Accuracy on 50 Random train.csv Questions

The official generation pipeline is:

- Index B
- dense retrieval `top_k=20`
- cross-encoder re-ranking to top-3
- Ollama `llama3`

Metric	**Value**
Official accuracy	70.00%
Correct answers	35 / 50

This is the formal HW3 score reported in the notebook and report.

6 Latency Analysis

Latency is measured on the same official pipeline and evaluation set.

Stage	Average Latency (s)
Vector search	0.0123
Re-ranking	0.0156
LLM generation	2.3950
Total	2.4229

Re-ranking adds about 15.6 milliseconds on average, which is very small compared with the 2.395-second LLM generation stage. On this rerun, proxy hit rate improves on Index A and ties on Index B, so re-ranking remains a low-cost stage that is operationally worthwhile in this pipeline.

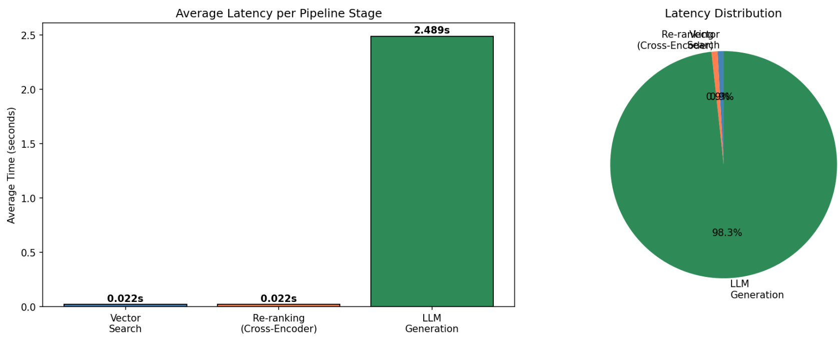

Figure 3: Latency breakdown from the official notebook run.

7 Optional Advanced Experiment

7.1 Motivation

The official random 50-question `train.csv` sample remains the required HW3 setting. The appendix branch was tested only to study whether stronger retrieval, answer-aware evidence scoring, and model variants improve science-only QA.

7.2 Science-only evaluation setup

The advanced branch filtered the public mirror with science keywords, creating a science-only pool with the following traceable setup from `advanced_experiment_summary.json`:

Metric	Value
Public mirror rows	6,684
Science-only pool rows	1,469
Science dev rows	10
Science held-out test rows	50

This appendix-only setup is not the official HW3 evaluation because it does not use the required random 50-question `train.csv` sample.

7.3 Advanced retrieval changes

The advanced notebook branch (`hw3advance.ipynb`) implemented:

- query rewriting for Wikipedia retrieval,
- long-form Wikipedia evidence extraction,
- chunk-level re-ranking over fetched evidence,
- option-aware evidence scoring,
- NLI-style scoring with a DeBERTa entailment model.

7.4 Model and ensemble comparison

The following advanced appendix numbers are traceable from `advanced_experiment_summary.json` and `advanced_rag_results.csv`:

Appendix experiment	Accuracy
Llama 3 model-only on corrected science dev slice	30.00%
DeepSeek model-only on corrected science dev slice	40.00%
DeepSeek RAG vector-only	30.00%
DeepSeek RAG with re-ranking	60.00%
Option-aware rerank + NLI	60.00%
Final ensemble on 10-question dev slice	80.00%
Final ensemble on held-out 50 science-only questions	72.00%
Post-hoc best rule on the same science-only run	74.00%

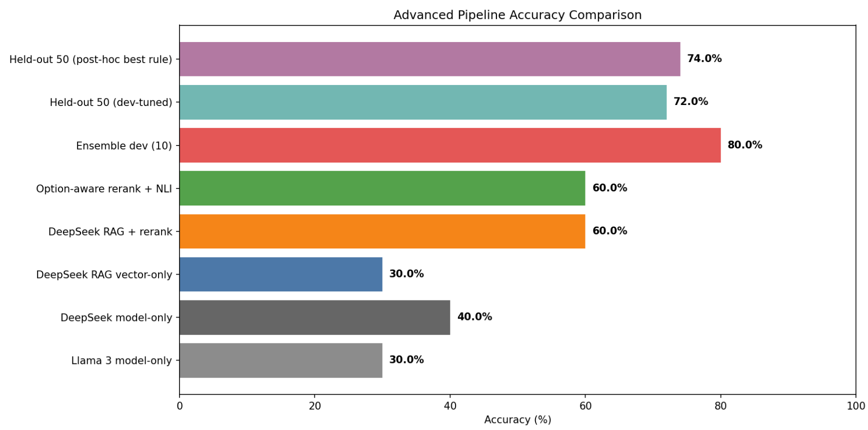

Figure 4: Appendix-only advanced method comparison from notebook outputs.

7.5 Why this is not the official HW3 score

These results are useful as an ablation and extension appendix, but they are not reported as the formal HW3 score because:

- they use a filtered science-only pool rather than the required random 50-question `train.csv` sample,
- they use DeepSeek and ensemble extensions beyond the official Ollama `llama3` pipeline,
- the formal HW3 result in this report remains the official **70.00%** score on the random-`train.csv` evaluation.

8 Conclusion

The final official HW3 pipeline uses a random 50-question sample from the specified Kaggle `train.csv` path, two chunking strategies, FAISS retrieval, cross-encoder re-ranking, and Ollama `llama3` generation on GPU. The final official accuracy is **70.00% (35/50)**. Strategy B gives the strongest vector-only hit-rate proxy, while the final official generation pipeline uses Strategy B with re-ranking as the required two-stage retrieval setup. The appendix shows that a science-only DeepSeek/ensemble branch can reach a higher ceiling, but the formal HW3 score reported here remains 70.00%.